

Dudilig Product Requirements Document v1.2.4

Version: 1.2.4 (May 2026, infrastructure-layer competitive update and co-founder scope parity patch) **Status:** Confidential. Supersedes v1.2.3.

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Source documents reconciled: Dudilig Bundled-Platforms Competitive Brief (May 2026); Dudilig Infrastructure-Layer Competitive Brief (May 2026); Ethereum Attestation Service docs; Quicknode EAS guide (Nov 2025); Chainlink Compliance Attestation guide (Feb 2026); Securitize S-4/A (April 2026); live homepage at www.dudilig.com; Anthropic Claude API documentation; Inspect AI evaluation framework; Stripe, Plaid, and Resend developer experience benchmarks.

What changed from v1.2.3 to v1.2.4

This is a focused patch. Every structural decision in v1.2.3 holds. v1.2.4 sharpens four things in light of the May 2026 bundled-platforms and infrastructure-layer competitive analyses, and brings the co-founder scope into explicit parity:

1. **Section 12: Competitive positioning rewritten as a three-tier map.** Tier 1 bundled platforms (Securitize, Apex+Tokeny+T-REX Ledger, Tokeny, Fireblocks). Tier 2 infrastructure attestation primitives and oracle/policy layer (EAS, Chainlink ACE/PoR/CRE). Tier 3 horizontal SaaS and bottoms-up data infrastructure (Vanta, Chainalysis, Persona). New explicit positioning: "the independent CCO workflow layer for tokenized funds, riding open attestation rails (EAS), independent of any issuance platform, oracle network, or chain." Includes a per-vector Chainlink ACE differentiation section and an 18-month watch list.
2. **Section 9 and Section 10.5: EAS as substrate, Chainlink as vendor-optional.** Build-vs-buy posture made explicit. Dudilig rides Ethereum Attestation Service (EAS) for cryptographic proof storage, immutability, and L2 portability. Chainlink CRE, Proof of Reserve, and Confidential Compute are integrated as vendor-optional inputs. The agentic stack stays oracle-neutral; no Chainlink dependency.
3. **Section 17: Co-founder scope rewritten for explicit parity.** Both Hersh Malik and Ken Tung Nguyen are building and shipping product across all frontiers. Lead-owner surfaces named for each founder, joint frontiers named separately, and complementary network and pedigree contributions documented. Both founders are on every shipping deliverable; the divide-and-conquer pattern is by surface, not by hours.
4. **Section 16: Acceptance criteria updated for v1.2.4 launch.** EAS schema registration milestone added. Public Trust Report URLs anchor to EAS. Section 17 parity rollout reflected in joint responsibility flags on capital formation and recruiting milestones.

Renumbering: none. Citations and sources remains Section 20.

1. Overview

Dudilig is the AI-native compliance work platform for tokenized real-world asset (RWA) issuers. It verifies that deployed smart contracts match off-chain compliance claims, automates continuous KYC and AML monitoring, and replaces human compliance services hours with AI agents priced per action.

Current state. Live marketing site at www.dudilig.com (Figma Sites build, polished, includes interactive pricing estimator and live evidence ticker with sample data). Demo prototype at dudilig.onrender.com (alias: demo.dudilig.com) with mock data throughout, except the Contract Analyzer surface, which calls the Claude API against real Solidity. Production features (live blockchain reads, real KYC integrations, public Trust Report URLs, Stripe-backed credit billing, multi-contract attestation) are the 30/60/90-day build target, contingent on the technical co-founder hire.

Category. New category: AI-native compliance work platform for tokenized assets. Not "compliance software" (Vanta), not "tokenization services" (Securitize, Tokeny), not "AI diligence" (Dili). Different category, different pricing, different acquirer multiple.

Reference shapes. Harvey for AI-native professional work priced per action.

Vanta for the three-sided compliance trust layer (issuer, customer, auditor).

Stripe for B2B2C horizontal expansion through embedded API and for developer experience.

Plaid for the developer-grade financial-data API surface.

Resend for the modern developer-first onboarding (sandbox first, no credit card required).

Lovable, Replit, and Cursor for usage-based engagement.

Structural promise. Dudilig is software, never a regulated financial services provider.

We do not custody assets, hold funds, register as a transfer agent, broker-dealer, or ATS, or provide investment, legal, or tax advice. Customers remain responsible for their regulatory obligations.

2. The problem

The tokenized RWA market reached approximately \$22 to \$26 billion on-chain in early 2026, up roughly fivefold in three years and concentrated in tokenized US Treasuries (BlackRock BUIDL, Franklin OnChain FOBXX, Hashnote USYC, Ondo OUSG). Source: RWA.xyz public dashboard, accessed Q1 2026 (excludes stablecoins).

Forward projections from credible institutions vary by an order of magnitude:

- McKinsey, "From ripples to waves: The transformational power of tokenizing assets" (June 2024): \$2 trillion base case by 2030, \$4 trillion bull case.
- BCG and ADDX, "Relevance of On-Chain Asset Tokenization in Times of Crypto Winter" (September 2022): \$16 trillion by 2030 in their primary scenario.
- Standard Chartered, "Real-World Asset Tokenization: From Ripples to Waves" (2024): approximately \$30 trillion by 2034.

Even the most conservative credible forecast implies roughly two orders of magnitude of growth from the current on-chain base.

The bottleneck blocking that growth is unit economics, not technology.

A Reg D 506(c) tokenized offering today carries an unbundled compliance cost stack of approximately \$300,000 to \$500,000 per launch when one allocates a full year of compliance overhead against a single offering. The line items: outside securities counsel for offering structuring and PPM (Cooley, Latham, DLA Piper class, sitting at the high end of their ranges because tokenization adds smart contract review and on-chain enforcement analysis); fund administration; KYC and identity vendor stack; sanctions screening; compliance operations headcount; smart contract security audit; and a Big Four annual audit when LP commitments require one. Source: bottoms-up estimate from public counsel rate cards and conversations with fund formation lawyers and tokenization platform operators.

The stack creates a hard floor on viable deal size. At \$300,000 of allocated compliance overhead, a \$5 million offering loses 6 percent of the raise. At \$50 million of raise size, the same overhead absorbs at roughly 0.6 percent, which is the practical break-even most boutique structurers identify. Below that threshold, tokenization is uneconomic.

Dudilig collapses that cost stack to under \$20,000 per offering using AI agents that perform the compliance work itself, denominated in usage credits, not seat licenses.

3. Customers (ICPs)

Provenance note. ICP ordering reflects a forward target composition modeled against analogous compliance markets (Vanta, Persona, Chainalysis) and informed by direct conversations with fund formation counsel and platform operators. Customer counts and revenue figures cited per ICP are 24-month modeled targets for design partner planning, not realized revenue. Validation against the open market begins with the design partner pilot.

ICP 1. Vertical B2B2C tokenization platforms (largest target revenue contributor)

Royal (music royalty tokenization), Lofty.ai (real estate), and the broader class of vertical platforms in sports memorabilia, collectibles, IP rights, agricultural assets, climate credits, fan tokens, and fractional luxury assets. Their end users are retail or prosumer; they need an embedded compliance backend their consumers do not see.

- Target profile: platforms with 50 to 500 underlying issuers where compliance cost is embedded and scales with end-user volume.
- Why structurally sticky: credit consumption scales with end-user volume, not seat count. Once the API is woven into onboarding, distribution, and on-chain enforcement, replacement requires re-architecting the consumer product.
- Displaceable spend per customer: \$1.5M to \$4.5M per year (in-house compliance ops, outside legal per jurisdictional launch, patchwork KYC vendor stack, internal tooling).

- Dudilig target capture per customer: \$1M to \$1.5M per year as credit consumption.
- Buying-signal language: "We can't onboard fans faster than our compliance ops team scales." "Our legal spend per jurisdictional launch is killing the consumer pricing model."
- 24-month target: 3 customers signed.

ICP 2. Tokenization-as-a-Service (TaaS) platforms (the design partner ICP)

Emerging and regional TaaS platforms whose own compliance stack is immature, who run on EVM, and who need third-party attestation feeds because their brand alone will not pass institutional diligence and Big Four audit. They are the layer underneath the Tokeny, Securitize, and ADDX tier.

- Confirmed and pursuable today: InvestaX (Singapore, MAS-licensed, Dudilig's v1 design partner). Chintai (Singapore). Brickken (Spain). Next wave: Allinfra (Hong Kong), Stobox, Tradeteq (UK), Hadron by Tether, and the Plume L2 ecosystem.
- Pursuit discipline: Tokeny, Securitize, Polymesh, and ADDX are competitor-or-channel, not direct prospects. They are pre-disposed to build or buy elsewhere.
- Why they buy: P&L is gated by compliance throughput. Dudilig replaces 60 to 80 percent of in-house compliance headcount cost. Third-party attestation feeds are more credible to institutional liquidity providers than self-issued ones.
- Displaceable spend per platform: \$500K to \$2M per year (3 to 8 in-house compliance FTE at \$150K to \$250K loaded, plus offering-by-offering external counsel).
- 24-month target: 11 customers signed.
- Cracking strategy: secure InvestaX as deeply engaged design partner in months 1 through 3. Use that logo to land 2 to 3 additional regional or vertical TaaS platforms by month 12. Open partnership conversations with Tokeny and Securitize as referral channels.

ICP 3. Tokenized fund managers, \$100M to \$2B AUM (the volume layer and cracking ICP)

Hashnote, OpenEden, Backed Finance, Maple, Goldfinch, Centrifuge issuers, plus the long tail of single-fund tokenization plays in Cayman, BVI, Singapore.

- Why this is the cracking ICP: shortest deal cycle (30 to 60 days). Cleanest v1 product surface (single issuer, single offering, Reg D 506(c)). Logo accumulation that unlocks every other ICP.
- Displaceable spend: \$200K to \$700K per fund per year.
- 24-month target: 67 customers signed.

- Buying-signal language: "We can't justify \$400K of legal for one offering." "Our fund admin can't read a smart contract."

ICP 4. Boutique investment banks and tokenization advisors

Mid-market IBs running tokenization advisory practices, regional broker-dealers, structured finance shops, family-office-affiliated advisory shops.

- Target profile: shops running 4 to 8 tokenized offerings per year seeking a repeatable deliverable.
- Sales motion: inbound from fund manager logos that recognize the work product and ask to license it for their own deal flow.
- Displaceable spend: \$300K to \$900K per shop per year.
- 24-month target: 22 customers signed.

ICP 5. HIP-3 deployers and RWA-perp market creators (strategic and PLG)

100+ active Hyperliquid HIP-3 deployers running RWA perp markets (NVDA, SpaceX, crude oil, equity indices), plus parallel deployers on Avantis, Ostium, Variational, Lighter.

- Target profile: low ACV, high strategic signal. Free tier expected to convert at the rate validated by analogous developer-tool funnels (target: 10 to 17 percent free-to-paid).
- Why this segment matters: strategic-acquirer signal (we are where on-chain RWA distribution is growing fastest), and the public Trust Report distribution surface that creates inbound demand from ICPs 1 through 3.
- 24-month target: 84 deployers active, of which approximately 14 paid.

Read-only stakeholders

Big Four auditors, internal compliance officers, and regulators access the read-only portal. Not a paying ICP at v1; they are the third side of the network whose adoption legitimizes the Trust Report.

4. Agentic architecture: the four pillars

Every production capability sits on one or more of these pillars. The pillars are non-negotiable. Any feature that cannot be defended on this spine should not ship.

Pillar 1. Multi-agent orchestration

A planner agent decomposes a compliance question into specialist sub-tasks. Specialists are narrow and tool-rich. Coordination happens through structured state in a shared workspace database, not by passing natural language between agents.

Specialist agents in v1:

- **Contract Attestation Agent.** Fetches deployed bytecode and source via RPC. Parses Solidity AST. Runs rule-by-rule checks against the selected standard. Emits structured attestation records. v2 treats the contract dependency graph (Token, IdentityRegistry, ComplianceModule, ClaimTopicsRegistry) as a first-class input.
- **Investor Monitoring Agent.** Event-driven. Watches KYC expiry, sanctions list deltas, PEP refresh cycles, accreditation re-verification. Writes alerts to the workspace.
- **Filings Agent.** Jurisdiction-aware deadline reasoner with calendar tools. Per-jurisdiction rule packs (CIMA, MAS, SEC, ADGM, BVI, SFC) drive what is tracked.
- **Rule Pack Agent.** Translates regulatory prose into testable assertions. Reg D 506(c), Reg S, MiCA, ERC-1400, ERC-3643 each become a versioned, machine-readable rule set.
- **Trust Report Agent.** Aggregates verified evidence into the public-URL artifact.
- **Chat Agent (RAG).** User-facing interface, scoped to a single workspace's vector store.

User-facing agents (the credit-consuming surface):

- **Issuance Agent.** Walks an issuer through structuring decisions for a new offering. 350 credits per offering.
- **Ongoing Compliance Agent.** Monitors the live offering continuously. 12 credits per investor per month.
- **Investor Q&A Agent.** Sits on the public Trust Center URL. 20 credits per question answered.

The determinism boundary (non-negotiable). LLMs synthesize and explain. They do not decide on facts. "Is this address on OFAC?" is a deterministic API call, not a Claude call. "Does this contract enforce a 250-investor cap?" is a static analysis result, not a model opinion. The model's job is to read those deterministic outputs and assemble the attestation, with citations.

Pillar 2. Proprietary training data (the data moat)

Five corpora that no general-purpose model has access to. Within twelve months of customer activity, this becomes Dudilig's structural defensibility.

1. **Contract-to-compliance mapping corpus.** Every analyzed contract paired with jurisdiction, claimed posture, and verified rule outcomes.
2. **Mismatch corpus (the gold dataset).** Cases where on-chain reality diverged from off-chain claims. The MAX_INVESTORS demo (500 deployed vs. 250 claimed) is the canonical shape.
3. **Attestation chain history.** Append-only, hashed. Captures contract evolution and compliance drift. Enables a temporal model: "this issuer's posture is degrading."

4. **KYC decision trails.** Why an investor was flagged, what was overridden, what was confirmed. Trains a risk scorer that beats off-the-shelf Persona or Onfido scoring.
5. **Cross-jurisdictional precedent.** What CIMA accepted, what MAS rejected, comment letter patterns. Few competitors will have this view across six jurisdictions simultaneously.

Rule packs themselves are a structured asset that compound with every customer-flagged false positive or missed issue.

Pillar 3. Evals (rigor at the level compliance demands)

Compliance is the domain where false negatives are legally actionable. The eval suite includes:

- **Golden contract set.** Curated bank with hand-labeled correct outcomes. T-REX reference, ERC-1400 reference, real Polygonscan deployments across passing and failing exemplars. Versioned.
- **Adversarial set.** Red-team contracts designed to evade detection: hidden backdoor mints, obfuscated transfer hooks, fake compliance modifiers, prompt-injection comments in Solidity. Non-negotiable.
- **Per-rule precision and recall.** Every compliance rule has its own metric. Investor-cap check needs near-100 percent precision; lockup-encoding has different tolerance.
- **Hallucination probes for the Chat Agent.** Ask about an investor that doesn't exist, ask for a hash that wasn't generated, ask about a jurisdiction we don't operate in. Any fabrication is a P0.
- **Cross-jurisdictional consistency.** Same contract analyzed under Reg D 506(c) vs. Reg S vs. MiCA should give consistent verdicts on overlapping rules.
- **Regression CI.** Every prompt change, model version bump, or rule pack edit runs the full suite. No silent quality drift.
- **Customer-specific golden sets.** Design partners get pinned evals using their actual contracts and investors. Their attestations are reproducible across model upgrades.
- **Multi-turn agent risk eval.** Methodology adapted from MT-AgentRisk (see section 11). Measure attack success rate (ASR) delta between single-turn and multi-turn prompts on every Chat Agent surface.

Numerical baselines, targets, and tooling for the eval suite are specified in section 18.

Pillar 4. Guardrails (where compliance products live or die)

Non-negotiable runtime constraints:

- **Schema-enforced JSON outputs.** Every verdict conforms to {rule_id, status, evidence, code_reference, model_version, rulepack_version, confidence}. Free-form compliance claims are rejected before they reach the user.
- **Citation requirement.** Every Pass or Fail must cite specific Solidity lines or rule-pack sections. Unsupported claims are blocked.
- **Confidence gating with human-in-the-loop.** Below threshold, the verdict goes to a review queue, not to auto-attestation. A compliance officer signs.
- **Tenant isolation in RAG.** Workspace-scoped retrieval, hard-enforced. No cross-tenant bleed even on shared embeddings.
- **Tamper-evident audit log.** Every agent action hashed and chained. Auditor mode can replay any decision with the exact model version, rule pack version, and inputs used.
- **Refusal patterns for legal advice.** Agent describes posture against a rule pack; does not opine on whether something is legal in a jurisdiction. That distinction is the difference between a tool and unauthorized practice of law.
- **Adversarial input sanitization.** Solidity comments containing prompt-injection patterns are stripped before model ingestion.
- **Reproducibility pinning.** Each attestation records (model_version, rulepack_version, RPC_block_height). Same attestation must be re-derivable months later.

Failure mode catalog and detection / recovery procedures per failure class are in section 18.7.

5. Core features (v1 scope, narrowed for pre-seed)

For v1.2.4 launch, only 5.3 (Smart Contract Attestation Engine) and 5.7 (Trust Certificate) ship to production. All others are built as prototypes for design partner feedback and move to v2.

5.1 Dashboard

Single-screen compliance posture view. Compliance health score (0 to 100); KPI cards (active investors, flagged issues, open filings, attestation coverage, AUM); token portfolio table; recent activity feed; upcoming filing deadlines. Current state: built, mock data. Pillar: presentation surface for outputs of all four pillars.

5.2 Investor Registry

Full KYC lifecycle management across all token holders. Investor list with KYC status (verified, pending, flagged, expired), AML risk score, sanctions screening timestamp, accreditation status; search and filter; investor detail drawer; alert surface for expired KYC, flagged investors, PEP matches. Current state: built, 15-investor mock dataset, no live KYC API connected. Pillars: Investor Monitoring Agent (orchestration); KYC decision trails (training data); guardrails (schema-enforced status fields).

5.3 Smart Contract Attestation Engine (THE UNIQUE IP)

Verify that on-chain contract behavior matches off-chain compliance claims. Step-by-step verification flow (fetch contract, parse transfer logic, check investor cap, verify jurisdiction whitelist, validate lockup, generate attestation record); per-rule Pass/Fail/Warning; mismatch simulation mode (toggle, surfaces MAX_INVESTORS 500 vs. 250 claim as the primary demo moment); attestation history table with result, fields checked, record hash. Current state: built as animated simulation. Production version reads live deployed contracts via Alchemy or QuickNode RPC and generates verifiable attestation records. Roadmap: replatform on the TxRay agent topology (planner, tool calls, executable self-check via semantic oracles) in days 31 to 60. Multi-contract dependency graph traversal in days 61 to 90. Pillars: orchestration; guardrails (schema-enforced verdicts, reproducibility pinning); evals (golden and adversarial contract sets).

5.4 Contract Analyzer

Live AI analysis of any Solidity contract against a selected compliance standard. Paste any Solidity source code; select standard (Reg D 506(c), Reg S, MiCA, ERC-1400, ERC-3643); output compliance score (0 to 100), overall status, Pass/Fail/Warning per rule with code references, critical issues list, ranked recommendations; copy report action. **Known limitation in v1:** T-REX-architecture tokens score 50 to 60 because compliance logic is delegated to external modules not included in single-contract analysis. Multi-contract analysis ships in days 61 to 90 (see section 11). Closed-source bytecode analysis path (productized from the IC3 Decompiling-with-LLM paper) expands addressable surface from the small minority of source-verified contracts to the full deployed contract universe. Current state: built and production-grade for source-verified contracts. Calls Claude API in real time. Pillars: orchestration (Contract Attestation Agent + Rule Pack Agent); training data (contract-to-compliance mapping corpus); guardrails (citation requirement on every verdict).

5.5 Regulatory Filings Tracker

Status management for all active regulatory filing obligations. Filing list across jurisdictions (CIMA, MAS, SEC, ADGM, BVI, SFC); status per filing (not started, in progress, submitted, overdue); days-until-due countdown; document checklist per filing; summary cards. Current state: built, 8-filing mock dataset across 6 jurisdictions. Pillars: Filings Agent (orchestration); cross-jurisdictional precedent (training data); rule pack registry (section 10).

5.6 Compliance AI Chat

Context-aware AI assistant with full workspace knowledge. Answers questions about investor KYC status, filing deadlines, attestation results, AML flags, audit risk; responses reference specific investor names, dollar amounts, contract details, recommended actions. Production version routes through Claude API. Hardening: multi-turn ASR eval (MT-AgentRisk methodology) on every Chat Agent surface. ToolShield-pattern runtime defense applied. Hallucination probes mandatory in regression CI. Current state: built, structured canned responses for the six highest-value queries. Pillars: Chat Agent

(orchestration, RAG with tenant isolation); evals (multi-turn methodology); guardrails (refusal patterns, hallucination probes).

5.7 Trust Certificate (renamed Trust Report in homepage copy)

Shareable compliance record for external parties. Seven verified compliance fields; attestation record with hash; plain-language summary for non-technical LPs; copy-link and download-PDF actions; cryptographic signature. Current state: built as in-app display. Production version generates a public URL accessible without login at [dudilig.com/trust/\[issuer\]/\[token\]](https://dudilig.com/trust/[issuer]/[token]). Privacy-preserving aggregate receipts (Verifiable Aggregate Receipts pattern, section 11) for quantitative claims: constant-size receipts that prove "N attestations were run" without disclosing which investors or contracts. Pillars: Trust Report Agent (orchestration); guardrails (cryptographic signing, tamper-evidence).

5.8 Onboarding Wizard

Guided setup for new issuers. Step 1: entity name, jurisdiction, contact. Step 2: token name, standard, network, contract address. Step 3: compliance toggles (accredited only, AML screening, sanctions monitoring). Step 4: first attestation result. Current state: built, flows to mock attestation result. Pillars: Issuance Agent (orchestration).

5.9 Compliance Credit Meter

Pricing primitive surfaced throughout the product. Remaining credits, usage bar, monthly allocation; tier label; top-up CTA. Current state: built, static display, no billing infrastructure connected. Stripe wired in days 31 to 60.

5.10 Auditor read-only portal

External-stakeholder evidence portal. Invite-link onboarding; read-only access to evidence streams, attestation log, audit-log replay; downloadable evidence packets (100-credit unit); standardized formats compatible with Big Four workflows. Pillars: Trust Report Agent + tamper-evident audit log (guardrails).

5.11 Dudilig Trust Center on Dudilig (eat-your-own-dog-food)

Dudilig's own compliance posture published through the same product surface paying customers use. The strongest credibility signal a compliance vendor can ship. Vanta uses the same move to anchor its own credibility.

Requirements:

- Public URL at dudilig.com/trust/dudilig, accessible without login. Same URL pattern as customer Trust Reports.
- Published evidence streams, each pulled live from the underlying source: SOC 2 Type I and Type II progress (live status pulled from Vanta API), ISO 27001 status, sub-processor registry with current DPA dates and per-vendor SOC 2 status, model versions currently in production with last-update timestamp, attestation lineage for

every attestation Dudilig has ever issued (privacy-redacted), security incident log if any, last penetration test date and remediation status.

- Tamper-evident audit log of the Trust Center's own update events, hash-chained.
- Cryptographic signature on every published artifact, identical to the customer Trust Report signing pipeline.
- Refresh cadence: every 24 hours via scheduled job, with an on-demand refresh button for internal users.
- Auditor read-only portal access (5.10) opened to Dudilig's own SOC 2 auditor as a working example for prospective customer auditors.

Implementation. The same Trust Report Agent that produces customer Trust Reports produces Dudilig's own. Same evidence streams, same signing pipeline, same agent path. Engineering cost is near-zero because the product is built for this. Production target: month 9, gated by SOC 2 Type I attestation arriving from Vanta (the first attestation worth publishing). A static placeholder version ships earlier (target: T+30) with the page structure in place and "in progress" indicators.

Pillars: Trust Report Agent (orchestration); tamper-evident audit log + cryptographic signing (guardrails); reproducibility pinning.

6. Pricing model

Usage-based, never seat-based. Modeled after Harvey, Lovable, Replit, Cursor.

Pricing primitive: Dudilig Compliance Credits. Every meaningful compliance action consumes credits.

Action	Credits	Effective price
Wallet sanctions screen (1 month of monitoring)	0.1	\$0.03 to \$0.10
Investor KYC verification	2	\$0.60 to \$2.00
Smart contract attestation event	5	\$1.50 to \$5.00
LP DDQ question answered	20	\$6 to \$20
Form D draft generation	50	\$15 to \$50
Auditor evidence packet	100	\$30 to \$100
Multi-jurisdiction analysis	200	\$60 to \$200
Issuance Agent full offering setup	350	\$105 to \$350

Effective per-credit price: \$0.10 to \$0.40 depending on tier and volume commitment. Target gross margin at scale: 80 percent or higher (AI inference cost target: 17 percent of revenue or lower). Per-action AI inference cost breakdown is in section 18.3.

Tier structure (six tiers).

Tier	Monthly credits	Bundle price	Effective \$/credit	Target ICP
Free	100	\$0	\$0	HIP-3 deployers, exploratory issuers, all developers (no credit card)
Issuer	1K to 5K	\$500 to \$2K	~\$0.40	Tokenized fund managers (ICP 3)
Boutique	5K to 15K	\$3K to \$7K	~\$0.30	Boutique IBs (ICP 4)
Platform	20K to 100K	\$5K to \$25K	~\$0.25	TaaS platforms (ICP 2)
B2B2C	200K to 500K	\$60K to \$150K	~\$0.20	Vertical B2B2C platforms (ICP 1)
Enterprise	500K+	\$50K to \$300K+	~\$0.10	BlackRock-class compliance teams or external auditors

Above the bundle, overage is priced at the tier's effective rate. Credits do not expire within the contract period.

Why usage-based, not SaaS. Compliance work is inherently variable. Issuers do not have stable headcount of compliance users to license; they have variable issuance volume, variable investor counts, variable regulatory events. Usage-based aligns price with the actual driver. Buyer psychology in finance is outcome-based: "\$3 per investor onboarded" lands; "\$48,000 per year per offering for unlimited seats" does not.

7. 30 / 60 / 90-day build targets

Time origin. All windows are measured from T+0 = technical co-founder start date. Not from PRD publish date. The CTO hire is the gating event.

Days 1 to 30: production foundation

- Live contract reads from Polygon and Ethereum via Alchemy or QuickNode RPC.
- Real attestation record generation with verifiable hashes.
- Public Trust Report URLs (no-login access).

- Stripe billing wired to credit meter.
- Persona KYC integration live.
- Compliance AI Chat routed through Claude API.
- Schema-enforced output contracts on every agent endpoint.
- Reproducibility pinning fields (model_version, rulepack_version, RPC_block_height) recorded on every attestation.
- Tamper-evident audit log (hash-chained agent actions) live.
- Tenant-isolated RAG implementation.
- Static Trust Center on Dudilig page live (placeholder evidence streams, "in progress" indicators).
- Inspect AI eval harness scaffolded; first 50-contract golden set hand-labeled and committed to repo.
- Langfuse or equivalent LLM observability deployed; every Claude call traced.
- sandbox.dudilig.com endpoint live with sk_test_ API keys.
- Design partner onboarded (target: InvestaX).

Days 31 to 60: agentic core, adversarial-safe Chat, developer surface

- Contract Attestation Engine replatformed on the TxRay topology (planner + specialist tool-calls + executable self-check via semantic oracles).
- MT-AgentRisk + ToolShield methodology adopted as the eval suite and runtime defense for the Compliance AI Chat and Investor Q&A Agent.
- Per-rule precision and recall harness operational across all rule packs; baseline numbers published internally.
- Hallucination probes added to regression CI.
- TypeScript SDK published to npm with full v1 API coverage.
- docs.dudilig.com live with quickstart, API reference (auto-generated from OpenAPI), and 5 example recipes.
- status.dudilig.com live.
- Customer target by end of window: 1 design partner in active pilot (InvestaX); 2 to 3 fund manager LOIs in late-stage diligence.

Days 61 to 90: institutional credibility, expanded reach

- **Multi-contract attestation traversal for T-REX / ERC-3643 architecture.** Token, IdentityRegistry, ComplianceModule, ClaimTopicsRegistry treated as a first-class graph input. This is the highest-priority lift in the 60 to 90 window because institutional tokenization runs on this architecture.
- **Closed-source bytecode analysis path** in the Contract Analyzer. Productized from the IC3 Decompiling-with-LLM paper. Expands addressable surface from source-verified to the full deployed contract universe. Required for ICP 5 (HIP-3) and the auditor portal.
- **IMMACULATE-style verifiable computation** wired into attestation record generation. Cryptographically detects model substitution and quantization abuse at low throughput overhead. Required for the Enterprise tier and auditor adoption.
- Structured rule packs shipped: Reg D 506(c), Reg S, ERC-3643. Each as a versioned database object with diffs, ownership, review workflow.
- dudilig CLI published to npm. Reg D 506(c) rule pack open-sourced under Apache 2.0 at github.com/dudilig.
- Per-attestation reasoning trace UI shipped (customer-facing transparency).
- Design partner case study published.
- Customer target by end of window: 3 to 5 paying customers signed (the design partner converted to paid, plus 2 to 4 fund managers).

Customer count discipline. Pre-seed customer counts are realistic, not aspirational. A fully-built engineering team of 2 to 3 produces this trajectory. Anything more aggressive without a CTO hire is a fiction, and we will not commit to it.

8. Roadmap beyond 90 days

Days 91 to 180 (months 4 to 6 of MVP)

- Resilient Alerting Protocols methodology adopted for the sanctions, KYC, and mismatch alert surface (section 11). Provides a defensible adversarial-cost property for the volumetric monitoring layer.
- First B2B2C platform customer signed (ICP 1, custom integration sprint).
- Python SDK published.
- 8 to 12 paying customers in total.
- SOC 2 Type I attestation achieved (via Vanta).
- Verifiable Aggregate Receipts (VAR) pattern shipped into the Trust Certificate. Privacy-preserving counts unlock auditor portal adoption in EU and HK markets.

Months 6 to 12

- Constraint-Friendly Map-to-Curve integration prototype for ZK-attestable verdicts. Cryptography work scoped contingent on cryptographer-grade hire or IC3 collaborator engagement. Not committed without that resource.
- 25 to 40 paying customers. Crossing \$2M to \$3M ARR run-rate as the realistic case; \$4M ARR is the bull case.
- First Boutique IB customer signed via inbound referral from fund manager logos (ICP 4).
- Compliance AI Chat shipped to all paying tiers.
- SOC 2 Type II attestation initiated.

Months 12 to 24

- Trajectory toward the modeled 24-month target composition: 3 vertical B2B2C platforms (ICP 1), 11 TaaS platforms (ICP 2), 67 fund managers (ICP 3), 22 boutique IBs (ICP 4), and a HIP-3 free-tier funnel. **These are modeled targets for planning, not committed forecasts.** Validation begins post-design-partner.
- Strategic acquirer conversations open.

v3 (post-acquisition independence path or future strategic surface)

- OML integration for on-prem fine-tuned compliance models at BlackRock-class enterprise customers.
- Cross-chain attestation aggregation.
- B2B2C embedded API surface formalized as a standalone SDK.
- White-label IB deployment.

9. Technical stack

Layer

Frontend

Backend

AI

Model gateway

LLM observability

Eval framework

Vector DB

Technology

Next.js 16, TypeScript, Tailwind CSS, shadcn/ui

Supabase (Postgres, auth, storage)

Claude API (Sonnet 4.6 for high-volume parsing, Opus 4.6 for complex legal analysis)

Portkey (multi-provider redundancy, cost monitoring, rate limit handling)

Langfuse (self-hosted)

Inspect AI (Anthropic-developed)

pgvector on Supabase

Prompt registry	Git-tracked YAML in monorepo (versioned with code)
Feature flags	GrowthBook (self-hosted)
Smart contract reads	Alchemy or Infura, ethers.js, ABI parsing
Attestation substrate	Ethereum Attestation Service (EAS) on Base (default), with Optimism, Polygon, and Ethereum mainnet anchoring options. EIP-712 signing. Merkle-tree private data for selective disclosure.
Oracle integrations	Chainlink CRE, Proof of Reserve, and Confidential Compute available as vendor-optional inputs. Stack stays oracle-neutral.
Smart contract attestation	Dudilig schema registered on EAS (one schema per rule pack version); attestation records issued via EAS attest function
KYC integrations	Persona, Sumsu, Parallel Markets
Sanctions screening	Chainalysis, Elliptic
Filings data	EDGAR scraper
Hosting	Vercel + Supabase + Railway
Auth	Clerk or Supabase Auth
Billing	Stripe
Eval and regression CI	Inspect AI + custom golden + adversarial sets, gated in CI
Audit log	Hash-chained Postgres table with replay tooling
APM	OpenTelemetry + Grafana Cloud
Error tracking	Sentry
Status page	Statuspage.io (statuspage.io to status.dudilig.com via CNAME)
Docs	Markdown source on GitHub, hosted on Vercel as docs.dudilig.com

Engineering principles. Every feature must be a credit-consuming action. Time-to-first-credit-consumed is the core activation metric (target: under 15 minutes from signup). Cost-per-credit needs ruthless engineering attention with a target gross margin of 80 percent or higher by month 18. No browser storage (localStorage, sessionStorage) anywhere; Postgres only. The Smart Contract Attestation Engine is the IP and patent-eligible work; document everything. Every Claude API call is traced through Langfuse with tenant tag and rule pack version. Every prompt change runs the Inspect AI eval suite in CI before merge.

Build-vs-buy posture. Ride public-good attestation rails (Ethereum Attestation Service) for cryptographic proof storage, immutability, and cross-L2 portability. Building a proprietary attestation primitive offers no defensible advantage over EAS, costs scarce engineering resources, and cuts Dudilig off from the Coinbase Verifications, Optimism, Base, and Polygon ecosystems already reading EAS. Integrate Chainlink CRE, Proof of Reserve, and Confidential Compute as vendor options where customer-driven, but never build the agentic stack as a dependency on any single oracle network.

The moat is the workflow, the cross-jurisdictional rule pack, and the mismatch corpus, not oracle infrastructure or attestation primitives.

10. First-class architectural objects (promoted from v1.0 implicit)

10.1 Rule pack registry

Each compliance standard (Reg D 506(c), Reg S, MiCA, ERC-1400, ERC-3643) is a versioned database object, not a dropdown selector. Requirements: version field; ownership and reviewer fields; diff view between versions; regression eval results required before promotion to production; subscription model so customer attestations are pinned to a specific rule pack version; deprecation lifecycle.

Why first-class: rule packs compound in value as customers flag false positives and missed issues. Without versioning, every fix risks invalidating prior attestations. With versioning, rule packs become a structured asset on Dudilig's balance sheet.

10.2 Attestation reproducibility pinning

Every attestation record captures (model_version, rulepack_version, RPC_block_height) at the moment of generation. The same attestation must be re-derivable months later by re-running with the same pinned inputs.

Why first-class: this is what makes the Trust Report regulator-credible. Without it, an auditor cannot verify that an attestation was correctly generated months after the fact. With it, Dudilig's output is admissible as workpaper evidence in a Big Four audit. Coupled with IMMACULATE-style verifiable computation in days 61 to 90, the model_version field becomes cryptographically attested rather than self-reported.

10.3 Multi-contract traversal (T-REX / ERC-3643)

The Contract Attestation Agent must treat the contract dependency graph as a first-class input. For ERC-3643 and similar architectures: Token, IdentityRegistry, ComplianceModule, ClaimTopicsRegistry. Per-module rules.

Why first-class: real institutional tokenization is mostly ERC-3643 / T-REX. The known-limitation flag in v1.0 (T-REX tokens scoring 50 to 60 due to single-contract analysis) is the single biggest gap blocking institutional credibility. Multi-contract traversal is the prerequisite for institutional ICPs, not a feature.

10.4 Tamper-evident audit log

Every agent action hashed and chained. Auditor mode replays any decision with the exact model version, rule pack version, and inputs used.

Why first-class: the auditor side of the three-sided network depends on this. Without replayability, the read-only auditor portal is a nice screenshot; with it, the portal becomes admissible evidence.

10.5 EAS attestation substrate posture

Dudilig's cryptographic attestation records are issued via Ethereum Attestation Service (EAS) schemas, not a Dudilig-proprietary attestation contract. Each attestation is a signed claim conforming to a registered Dudilig schema (one schema per rule pack version). On-chain attestations are stored on Base by default, with optional Optimism, Polygon, and Ethereum mainnet anchoring. Off-chain attestations use EAS off-chain mode with Merkle-tree private data when investor-level information must remain confidential. EAS Scan is the public verification surface external auditors use to inspect any Dudilig-issued attestation.

Why first-class: EAS is open public-good infrastructure adopted by Coinbase Verifications, Optimism, Base, Polygon, and identity protocols. Riding it gives Dudilig free immutability, free cross-L2 portability, and free interoperability with the broader attestation ecosystem. Reinventing this layer would consume scarce engineering capacity that should go into the workflow, the rule pack, and the agentic core. This is the build-versus-buy decision that makes the workflow moat the only thing engineering builds.

Pillars: orchestration (Trust Report Agent emits via EAS); guardrails (cryptographic signing pipeline pinned to EAS schema UID and EIP-712 signer); reproducibility pinning extended with EAS attestation UID at the moment of issuance.

11. IC3 research integration (the productization map)

Five papers from the Initiative for Cryptocurrencies and Contracts (IC3, co-headquartered at Cornell Tech) are slotted into the 30/60/90 build targets. Each is paired with the product capability it unlocks.

#	Paper	Roadmap window	What it unlocks
1	TxRay (Wang, Yu, Qin, Song, Gervais, Zhou; arxiv preprint 2602.01317)	Days 31 to 60	Architectural prerequisite. Contract Attestation Engine replatformed on planner + tool-calls + executable self-check pattern.
2	Decompiling Smart Contracts with an	Days 61 to 90	Closed-source bytecode analysis

	LLM (David, Zhou, Song, Gervais, Qin; arxiv preprint 2506.19624)		path. Expands addressable surface dramatically.
3	IMMACULATE (Guo, Qu, Wu, Song, Zhang et al.; arxiv preprint 2602.22700)	Days 61 to 90	Cryptographic backbone of reproducibility pinning. Required for Enterprise tier and auditor adoption.
4	Resilient Alerting Protocols (Mouallem, Breidenbach, Eyal, Juels; arxiv preprint 2602.10892)	Days 91 to 180 (v2)	Adversarial-cost property for sanctions monitoring. Relevant at B2B2C scale (50K+ wallets).
5	MT-AgentRisk + ToolShield (Li, Yu, Pan et al.; arxiv preprint 2602.13379)	Days 31 to 60	Multi-turn safety methodology and runtime defense for the Investor Q&A Agent and Compliance AI Chat.

Citation discipline. All arxiv IDs above are recorded as the references currently held in the IC3 thread export. Pre-publication, the engineering team must verify each ID and author list against the live arxiv listing and update if the preprint has moved. Do not cite externally without that verification step.

Tier 2 references queued: Verifiable Aggregate Receipts (eprint 2025/2330); Constraint-Friendly Map-to-Curve (eprint 2025/1503); OML (arxiv 2411.03887); Marino-Juels position paper (arxiv 2507.08249).

Honest dependency. The largest target revenue line in the company (B2B2C, ICP 1) depends on the days 61 to 90 work shipping, particularly multi-contract traversal and the bytecode analysis path. If those slip, ICP 1 economics slip with them.

12. Competitive positioning (the wedge)

Dudilig is **the independent CCO workflow layer for tokenized funds**: the application layer that rides open attestation rails, sits independent of any issuance platform, oracle network, or chain, and replaces human compliance services hours with AI agents. Independent does not mean neutral. Independent means software-only, with no conflicts of interest across issuance, custody, transfer agency, broker-dealer, fund admin, or oracle infrastructure. Dudilig is the only player in the tokenization compliance market that can credibly claim that.

12.1 The three-tier map

The tokenization compliance market resolves into three tiers. Each has a different set of incumbents, a different buyer, and a different defensible moat. Conflating them is what lets a competitor sound bigger than they are.

Tier 1: Bundled platforms (issuance, transfer agency, fund admin).

- **Securitize.** Vertically integrated regulated services (transfer agent + broker-dealer + ATS + fund admin). Going public via Cantor Equity Partners II SPAC at approximately \$1.25B, revenue +841 percent in 2025. Cannot be a neutral horizontal layer because issuers on competing platforms will not feed compliance data into a competing issuance platform.
- **Apex Group + Tokeny + T-REX Ledger.** Apex (fund admin on \$2T+ AUA) plus Tokeny (issuance platform, ERC-3643 standard owner) plus T-REX Ledger (cross-chain on-chain compliance state, launched March 2026). Closer to a horizontal claim than Securitize, but conflicted across issuance platforms (Tokeny competes with other issuers; Apex competes with other admins) and ERC-3643-centric. European institutional center of gravity.
- **Tokeny standalone.** EU per-issuer SaaS. T-REX standard owner.
- **Fireblocks / Anchorage.** Custody-bundled tokenization, per-asset fees.

Dudilig stance: not direct competitors at the workflow layer. Plausible distribution partners (Securitize ID API surface, Apex fund-admin client base) and plausible long-term acquirers.

Tier 2: Infrastructure attestation primitives and oracle / policy layer.

- **EAS (Ethereum Attestation Service).** Public-good substrate. Two smart contracts (schema registry plus attestation contract) live on Ethereum, Optimism, Base, Polygon. Free. EIP-712 signing. Merkle-tree private data. No buyer; adopted by Coinbase Verifications, Optimism, Base, identity apps.
- **Chainlink.** Industry-standard oracle network plus compliance modules: Automated Compliance Engine (ACE) for on-chain rule enforcement, Proof of Reserve, Chainlink Runtime Environment (CRE) for custom data, Confidential Compute (proprietary ZKP), Privacy Standard, Digital Transfer Agent (DTA), DataLink. Distribution via Swift, JPMorgan, BUIDL, ANZ relationships.

Dudilig stance: ride EAS for proof storage, immutability, and L2 portability (section 10.5). Free, neutral, already adopted. Compete with Chainlink ACE on a different vector (section 12.3). Stay oracle-neutral on the agentic stack so Chainlink can be integrated as a vendor option without becoming a dependency.

Tier 3: Horizontal SaaS compliance and bottoms-up data infrastructure.

- **Vanta / Drata / Secureframe.** Horizontal SaaS compliance (SOC 2, ISO). Per-seat ARR. Different category. Vanta is a structural reference and a Layer 1 input to Dudilig's own posture (section 13).
- **Chainalysis / Persona / Elliptic.** Bottoms-up specialized infrastructure. APIs. Different layer. Dudilig integrates them as data sources and owns the workflow above.

Dudilig stance: not competitors. Vendor inputs to the workflow layer. Vanta is the natural acquirer if it decides to enter the tokenized-asset category.

12.2 Why Dudilig is the only independent application layer

Independence is a positioning claim that has to be backed by structure. In Dudilig's case it is: software-only, no issuance platform, no custody, no transfer agency, no broker-dealer, no fund admin, no oracle network, no chain affiliation. Apex+Tokeny cannot claim this (fund admin plus issuance platform plus chain layer). Securitize cannot claim this (transfer agent plus broker-dealer plus ATS plus fund admin). Chainlink cannot claim this in the same way (oracle network with proprietary compliance modules and token economics). EAS does not have a workflow at all; it is plumbing.

Three compounding advantages, all execution-dependent:

1. Software-only, zero conflicts of interest across issuance, custody, transfer agency, broker-dealer, fund admin, oracle infrastructure, or chain affiliation.
2. Cross-jurisdictional rule pack and mismatch corpus that compound with attestation volume.
3. CCO-workflow-first design across any token standard, any chain, any issuance platform, any oracle network.

None are absolute. All require fast, focused execution.

12.3 Differentiation against Chainlink ACE specifically

Chainlink is the closest competitor on the policy / workflow vector. The defensible delta is real but specific.

- **Buyer.** ACE sells to protocols and issuer engineering teams building smart contracts. Dudilig sells to CCOs at fund managers who think in audit cycles, evidence packs, and regulator-readable language. ACE does not speak that.
- **Scope.** ACE is an on-chain policy engine. Compliance is roughly 90 percent off-chain workflow: auditor email threads, attorney memos, LP trust reports, regulator filings, Slack pings on drift. Dudilig owns that surface; ACE does not.
- **Posture.** ACE is rules-engine-first, oracle-first. Dudilig is workflow-first, agentic-first.

The risk to take seriously: Chainlink ships a CCO-facing UI on top of ACE in the next 18 to 24 months. They have capital and customer base. Dudilig must move faster, own the rule pack

and corpus before they catch up, and remain oracle-neutral so the agentic stack does not become a Chainlink dependency.

12.4 Disciplined positioning rule

If Dudilig ever drifts into "we are Securitize but cheaper" or "we are Vanta for crypto" or "we compete with Chainlink on oracle infrastructure," the wedge collapses. Exact framing: "the independent CCO workflow layer for tokenized funds, riding open attestation rails (EAS), independent of any issuance platform, oracle network, or chain." Anything broader and the category swallows us.

12.5 Watch list (next 18 months)

1. Whether Chainlink ships a CCO-facing UI on top of ACE. If it does, the differentiation collapses to buyer plus workflow depth plus rule pack. Track Chainlink compliance-product hiring as the leading indicator.
2. Whether Apex Group plus Tokeny integrate Chainlink ACE into T-REX Ledger. That combination would compress Tier 1 and Tier 2 into a single bundled stack and accelerate the timeline to compete.
3. Whether EAS adds an opinionated compliance schema library that Dudilig either contributes to (preferred) or routes around.
4. Whether Securitize or its post-SPAC entity ships an in-house "Securitize Verify" lookalike to lock single-platform issuers before Dudilig can pitch.
5. Whether a credible APAC issuance platform attempts a horizontal compliance positioning that mimics Dudilig's wedge.

6. Compliance posture (Dudilig's own): the six-layer stack

A vendor whose product output is a cryptographically signed compliance attestation is sold against risks Vanta does not cover. Vanta certifies security and process controls (SOC 2, ISO 27001, GDPR procedural shell). It does not cover Dudilig's own AML and sanctions program, model risk management, attestation product liability, vendor risk management for the sub-processor chain, examination readiness for bank-aligned customers, or AI-specific regulatory compliance (Layer 6, new in v1.2.3). Vanta is layer 1 only.

Layer	Coverage	Tooling	Owner timing
1. Security and process	SOC 2 Type I and II, ISO 27001, GDPR procedural shell	Vanta	Vanta engaged day one. Type I by month 9, Type II by month 15, ISO 27001 by month 18 if EU pipeline warrants.

2. Privacy and data processing	GDPR Article 28 data-processor duties (DPA framework, sub-processor disclosure, 72-hour breach notification, data-subject request handling); CCPA and the 15-plus US state privacy laws now in effect; data residency for Singapore PDPA, UAE, EU sovereignty	DPA template, sub-processor registry, Vanta privacy module	Month 3. DPA template ships before the first paying customer.
3. Securities-adjacent compliance program (Dudilig's own)	KYB on every issuer customer; OFAC, UN, EU sanctions screening on customers and on Dudilig's own treasury wallets; UPL (unauthorized practice of law) enforcement at every Chat Agent surface; operational logging of the "no investment advice" guardrail	Persona for KYB; Chainalysis sanctions screen on Dudilig's own treasury wallets; internal program doc	Month 6. Before the first paying customer signs.
4. Model risk and product liability	SR 11-7-aligned model risk management framework; model versioning and reproducibility (section 10.2); attestation E&O insurance; contract liability cap framework	Internal MRM doc; insurance broker (Founder Shield, Vouch); standard contract template with capped indemnity	Month 9. Before the first Platform-tier deal.
5. Regulatory examination readiness	Vendor-examination workpaper packets; FFIEC vendor management	Internal evidence room; Vanta export bundles; outside	Month 15. Before the first bank-aligned customer.

	alignment; NYDFS Part 504 readiness for NY-licensed bank customers; state-by-state interpretation tracking on the moving line between "compliance software" and "service that looks like money transmission"	counsel quarterly review	
6. AI-specific regulatory compliance (NEW in v1.2.3)	EU AI Act high-risk system classification (financial compliance attestation likely qualifies); Colorado AI Act (effective 2026) and parallel state-level AI laws; GDPR Article 22 right-to-explanation when an investor is flagged or rejected; algorithmic bias and fairness assessment for KYC and accreditation models; published model cards per attestation; human oversight documentation per EU AI Act Article 14	Internal AI governance doc; published model cards on Trust Center (section 19.7); algorithmic bias audit pipeline; EU AI Act conformity assessment with notified body if classified high-risk	Month 12. Before EU MiCA-jurisdiction customer signs.

Implementation notes.

- **Vanta math.** Approximately \$60K to \$90K Year 1 with Vanta and audit fees, vs. approximately \$250K to \$350K to build the same posture in-house. Time-to-Type-I is 60 to 90 days with Vanta vs. 6 to 12 months without. Source: Vanta and audit-firm public pricing plus comparable vendor benchmarks. The slight irony of a compliance vendor running on Vanta is fine because the categories are different. None of that argues for using Vanta as the entire compliance posture, which is what the v1.0 framing implied.

- **Insurance program.** Tech E&O at \$5M minimum from month 9, raised to \$10M before the first Enterprise-tier deal. Cyber liability \$5M. Crime coverage \$1M. D&O \$2M at Series A. Without underwriters behind the contractual liability caps, the caps are not credible to the customer's general counsel.
- **Vendor risk management.** Dudilig is itself a sub-processor for issuer customers and depends on a sub-processor chain: Persona, Sumsb, Chainalysis, Elliptic, EDGAR scraper, Alchemy, Anthropic, Stripe, Vercel, Supabase, Clerk. Each requires a DPA, sub-processor disclosure to issuer customers, and an annual vendor risk assessment. By month 12, this is a discrete program with an owner.
- **Hiring timeline.** Fractional CCO on retainer by month 9, roughly 10 hours per month at approximately \$5K per month, sourced through Continuum, Compliance.ai, or a named ex-Big Four senior on retainer. First full-time compliance hire post-seed at month 15 to 18. Hire profile: ex-Big Four audit senior or ex-Securitize / ex-Tokeny compliance ops manager. Cost approximately \$180K to \$220K loaded.
- **AI governance lead (new in v1.2.3).** Layer 6 requires a named AI governance owner. At pre-seed and Series A, this is the CXO (Ken) with technical co-founder support. By Series B, a dedicated AI governance lead (or external counsel with AI specialization) is required. Candidates: ex-FTC Bureau of Consumer Protection, ex-NIST AI Risk Management Framework contributor, or an academic with EU AI Act conformity experience.

The signal multiplier. A Trust Center on Vanta is the floor signal. The strong signal for a compliance product company is to publish its own Trust Report on Dudilig itself at dudilig.com/trust/dudilig (see section 5.11): SOC 2 progress, sub-processor list, model versions in production, attestation lineage, all served from the same product surface paying customers use. Engineering cost is near-zero because the product already does it for issuer customers.

The Out-of-Scope-Forever line still holds. No SEC transfer agent registration, no FINRA broker-dealer license, no ATS registration, no state money transmitter licenses, no FinCEN MSB registration. That structural choice is what makes the AI-platform revenue framing legitimate. Layer 5 examination readiness exists precisely to defend the line as state and federal interpretations move. Layer 6 is the parallel work for AI-specific regulation as that landscape settles.

14. Out of Scope (V1)

- Reg S, Reg A, Reg CF support (v2)
- Multiple offering types per issuer (v2)
- Investor-facing trust center marketplace (v2)
- Cross-chain attestation aggregation (v2)
- HIP-3-deployer-specific tooling (v2)

- B2B2C embedded API surface (v2; first B2B2C deal still ships on a custom integration sprint)
- White-label deployment for IBs (v3)
- Filing submission (we track only, we do not submit)
- Non-EVM chains (deferred; revisit at v3)
- Go SDK (deferred to v2)
- GraphQL API (deferred to v2; REST plus webhooks at v1)

15. Out of Scope (Forever): the structural promise

These categories are out of scope permanently because crossing the line collapses the AI-platform revenue framing:

- No SEC transfer agent registration. Securitize and partners own that role.
- No FINRA broker-dealer license. We do not execute trades.
- No ATS registration. We do not operate a trading venue.
- No state money transmitter licenses. We do not move funds.
- No FinCEN MSB registration. We are not a money services business.
- No investment advice. Agent describes posture against rule packs; never opines on legality.
- No custody. We never hold customer funds or assets.
- No trading, marketplace, or secondary-market functionality.
- No cap table management (this is licensed transfer agent work).

This line is the structural decision that makes Dudilig software, not services. It is what makes the AI-platform revenue framing legitimate and the acquirer multiple high. Do not drift across it under any pressure.

16. Acceptance criteria for v1.2.4 launch

The v1.2.4 build (90-day window from CTO start date) ships when:

1. Contract Attestation Engine is replatformed on the TxRay agent topology and produces real attestations against live deployed contracts on Polygon and Ethereum, with reproducibility pinning fields populated and tamper-evident audit log enabled.

2. Multi-contract traversal for ERC-3643 / T-REX returns a coherent compliance score on the Tokeny T-REX reference and at least three real ERC-3643 deployments. T-REX tokens no longer score in the 50 to 60 range.
3. Closed-source bytecode analysis path returns a usable analysis on at least three unverified live HIP-3 deployer contracts.
4. Compliance AI Chat passes a multi-turn ASR eval (MT-AgentRisk-style) at the threshold defined in section 18.1, with ToolShield-pattern runtime defense active.
5. Stripe billing produces correct credit-meter behavior across at least the Free, Issuer, Boutique, and Platform tiers in production.
6. Persona KYC integration is live and producing real KYC decision trails into the workspace database.
7. Public Trust Report URLs render at [dudilig.com/trust/\[issuer\]/\[token\]](https://dudilig.com/trust/[issuer]/[token]) with cryptographic signing, plain-language summary, and a privacy-preserving aggregate receipt for any quantitative claim.
8. Auditor read-only portal is shippable end-to-end on at least the design partner's data, with audit-log replay working.
9. Per-rule precision and recall reports on the golden and adversarial set are stored and trended in CI; baseline numbers published internally per section 18.1.
10. SOC 2 Type I engagement with Vanta is open with a clear Type I target by month 9.
11. **Developer experience milestones (NEW in v1.2.3):** TypeScript SDK published to npm with full v1 API coverage; docs.dudilig.com live with quickstart (sub-10-minutes to first attestation), API reference auto-generated from OpenAPI, and 5 example recipes; sandbox.dudilig.com endpoint live with sk_test_ API keys and three pre-loaded demo contracts; status.dudilig.com live; Reg D 506(c) rule pack open-sourced under Apache 2.0 at github.com/dudilig.
12. **AI transparency milestones (NEW in v1.2.3):** Per-attestation reasoning trace UI shipped; eval transparency dashboard shipped (customers can see Dudilig's per-rule precision/recall); model card per attestation rendered.
13. 3 paying customers signed: target composition is 1 design-partner platform on a paid pilot (InvestaX) plus 2 fund managers, with the technical co-founder hired and the core attestation engine in production.
14. **EAS substrate milestone (NEW in v1.2.4):** Dudilig schema registered on Ethereum Attestation Service on Base, with parallel registrations on Optimism and Polygon. Public Trust Report URLs anchor every attestation to a verifiable EAS attestation UID viewable on EAS Scan. Reproducibility pinning extended to include the EAS attestation UID alongside (model_version, rulepack_version, RPC_block_height).

15. **Oracle-neutrality milestone (NEW in v1.2.4):** Chainlink CRE, Proof of Reserve, and Confidential Compute integrated as vendor-optional inputs behind a single oracle abstraction layer. No agentic-stack dependency on a single oracle network. The abstraction passes a swap test where the same attestation can be regenerated against an alternative oracle vendor with byte-equal verdict on overlapping rules.
16. **Capital and recruiting joint milestone (NEW in v1.2.4):** Pre-seed round closed with a co-led pitch from both founders. Technical co-founder hired through a parallel-pipeline process owned jointly by Hersh and Ken (section 17.5).

17. Co-founder scope and ownership

Two-founder shipping company. Both Hersh Malik and Ken Tung Nguyen are building and shipping product across all frontiers. The map below names the lead-owner for each surface, but both founders are operationally on every shipping deliverable. The divide-and-conquer pattern is by surface, not by hours: lead ownership clarifies decision rights; it does not ration contribution.

17.1 Hersh Malik (CEO) lead surfaces

- Product architecture and engineering direction. The four pillars (section 4), the agentic core, the eval suite, the determinism boundary (section 4.1), the schema-enforced output contracts.
- Regulatory positioning of the platform. The structural promise, the Out-of-Scope-Forever line (section 15), Layers 1 through 5 of the compliance stack (section 13).
- AI governance and Layer 6 conformance. EU AI Act high-risk system classification, Colorado AI Act, GDPR Article 22 right-to-explanation, model card publication, human oversight documentation.
- Vanta engagement, SOC 2 Type I and Type II path, ISO 27001 engagement when the EU pipeline warrants it.
- Customer-facing product surface for the attestation engine, the dashboard, the Trust Report, the auditor read-only portal.
- Developer experience program (section 19) and AI-native operations (section 18), in lockstep with the technical co-founder once hired.
- Pedigree and network: product and engineering operating leadership, regulatory and AI-governance fluency, Vanta and audit-firm relationships, builder community access, infrastructure-vendor relationships across the Anthropic, Inspect AI, Langfuse, Portkey, and GrowthBook stack.

17.2 Ken Tung Nguyen (CXO) lead surfaces

- Capital formation. Pre-seed, seed, strategic acquirer pathway. Investor narrative across the Bundled-Platforms Competitive Brief and the Infrastructure-Layer Competitive Brief (section 12 and section 12.1 to 12.5).

- Partner network and customer pipeline. APAC tokenization platforms with InvestaX as the v1 design partner, plus Chintai, Allinfra, the Plume L2 ecosystem; US and EU fund managers; family offices and boutique investment banks; fund administrators including pathway conversations with Apex Group on the non-conflicted segments of their book.
- IC3 and Cornell research collaboration pathway. Liyi Zhou, Ari Juels, Lin William Cong. The five-paper integration map (section 11) and the Tier 2 references queued in that section.
- GTM engineering. Pricing primitive (credits and tier structure, section 6), free-to-paid funnel mechanics, design-partner conversion sequence, Trust Report distribution flywheel, sandbox-to-paid conversion analytics.
- Competitive intelligence and infrastructure-layer positioning. The three-tier framing (section 12.1), the EAS substrate posture (section 10.5), the Chainlink ACE differentiation vector (section 12.3), the watch list (section 12.5).
- Boutique IB, fund manager, and family office channel ownership. Inbound from fund-manager logos that recognize the work product and ask to license it for their own deal flow (ICP 4 sales motion).
- Pedigree and network: finance and capital markets pedigree, AI product management discipline, APAC operator and licensed-platform relationships, US and Singapore fund-manager access, Cornell IC3 research network, founder-community access across the Anthropic and AI-native operator circles.

17.3 Joint frontiers (both founders ship)

- Design partner relationship. InvestaX or similar ICP is the v1 design partner; both founders are on every operating call.
- Pre-seed and seed capital narrative. Both pitch. Hersh leads the product and engineering deep-dive; Ken leads the GTM, ICP, and competitive-positioning narrative.
- Recruiting. Technical co-founder hire is the single biggest gating item; joint priority.
- Public positioning, conference and podcast surface, partnership conversations with Securitize, Tokeny, Apex Group, and the Chainlink BD organization.
- Trust Center on Dudilig publication (section 5.11) as the eat-your-own-dog-food credibility flywheel.
- Strategic acquirer conversations.

17.4 Functional ownership table

Area	Lead	Joint
Product architecture, agentic core, eval suite	Hersh	

Regulatory positioning (Layers 1 to 5) and AI governance (Layer 6)	Hersh	
Vanta, SOC 2, ISO 27001	Hersh	
Customer-facing product surface (attestation, dashboard, Trust Report, auditor portal)	Hersh	
AI-native operations (section 18) and developer experience (section 19)	Hersh, with technical co-founder	
Capital formation and investor narrative	Ken	
Partner network (APAC, US, EU fund managers, fund admins, IBs)	Ken	
IC3 research outreach (Liyi Zhou, Ari Juels, Lin William Cong)	Ken	
Competitive intelligence and infrastructure-layer positioning	Ken	
GTM engineering, pricing primitive, free-to-paid funnel	Ken	
Boutique IB and family office channel	Ken	
Design partner relationship (InvestaX)		Hersh + Ken
Pre-seed and seed pitch		Hersh + Ken
Technical co-founder recruiting		Hersh + Ken
Trust Center on Dudilig publication		Hersh + Ken
Strategic acquirer conversations		Hersh + Ken
Engineering, AI-native ops execution post-CTO start	Technical co-founder / CTO (OPEN, BLOCKING, 1 to 2 percent equity, 4-year vest, 1-year cliff)	

Developer relations and community (post-CTO hire) TBD by month 6

17.5 Critical open item

Technical co-founder or lead engineer hire is the single biggest gating item. No 30-day target ships without this person committed. Joint priority for both Hersh and Ken; both founders are running parallel recruiting pipelines into the same shortlist.

17.6 Network and pedigree complementarity

The two founders bring distinct, additive networks. Hersh's product, engineering, regulatory, and AI-governance pedigree pair with Ken's finance, capital markets, APAC operator, and IC3 research network. Neither network alone reaches the customer base or the cap table required for the 24-month modeled trajectory. Both founders are on every shipping deliverable; lead ownership clarifies decision rights, not contribution. The compounding effect is intentional: every customer conversation, every fundraising meeting, every research collaboration, and every recruiting pipeline benefits from both founders being current on the work.

18. AI-native operating metrics and tooling (NEW in v1.2.3)

Compliance products are scrutinized on numbers, not promises. This section commits Dudilig to specific baselines and targets, names the tooling that produces them, and catalogs the failure modes the product must handle.

18.1 Eval suite metrics: baselines and targets

Per-rule precision and recall targets. Baselines established by T+30 from CTO start with a 50-contract hand-labeled golden set. Targets achieved by T+90 with full Inspect AI eval suite operational.

Rule	Precision target	Recall target	Why
Investor cap (e.g., MAX_INVESTORS)	99.5%	99.0%	False negatives are legally actionable under Reg D 506(c).
Jurisdiction whitelist	99.5%	99.5%	Both directions matter; over-inclusion and under-inclusion are both audit findings.
Lockup encoding	99.0%	95.0%	False positives are tolerable; missed lockup is high-severity.

Transfer hooks (backdoor detection)	99.9%	90.0%	Backdoor false positive is expensive but a missed backdoor is catastrophic.
Sanctions sweep	99.9%	100.0%	OFAC compliance is a regulatory hard requirement. Missed match is a P0 incident.
Accreditation	99.0%	98.0%	Standard rigor for Reg D 506(c) accreditation verification.
Rule pack version pinning	100.0%	100.0%	Deterministic. Any deviation is a P0 bug.

Hallucination rate methodology. Adversarial probe set of 200 questions covering: investor-not-exists; hash-not-generated; jurisdiction-not-operated; rule-pack-version-not-deployed; contract-not-attested. Probe set runs weekly in CI. Target: under 0.1 percent fabrication rate. Baseline measurement: by T+30. P0 escalation: any single fabrication detected in production triggers immediate model rollback and incident report.

Multi-turn ASR (MT-AgentRisk methodology). Adversarial probe set of 100 multi-turn jailbreak attempts. Single-turn ASR baseline target: under 2 percent. Multi-turn ASR target: under 5 percent (the published MT-AgentRisk delta is +16 percent multi-turn vs single-turn). ToolShield runtime defense reduces ASR by approximately 30 percent per the paper. Target post-ToolShield ASR: under 3.5 percent.

Cross-jurisdictional consistency. For overlapping rules between Reg D 506(c), Reg S, and MiCA, the same contract analyzed under different rule packs must yield consistent verdicts on overlap. Target: 100 percent consistency on overlapping rules.

Customer-pinned eval reproducibility. Every customer-paid attestation is pinned to its (model_version, rulepack_version, RPC_block_height) triple and is re-derivable for at least 7 years (regulatory retention period). Reproducibility test runs monthly on a 10 percent sample of historical attestations.

18.2 Latency and reliability targets

Surface	P50	P95	P99
Contract Attestation Agent (single-contract, full attestation)	12 s	30 s	60 s

Contract Attestation Agent (multi-contract T-REX traversal)	30 s	90 s	180 s
Investor Q&A Agent (RAG response)	3 s	8 s	15 s
Issuance Agent (full setup wizard, end-to-end session)	2 min	5 min	10 min
Compliance AI Chat (first token)	2 s	5 s	10 s
Filings Agent (deadline reasoner)	5 s	15 s	30 s
Trust Report compile	8 s	20 s	45 s
Sanctions screen (deterministic API call)	200 ms	800 ms	2 s
KYC verification (Persona round-trip)	30 s	90 s	3 min

Reliability SLAs.

- Uptime: 99.9 percent for paid tiers (43 minutes downtime per month allowed). Free tier: best-effort.
- Error budget: 0.1 percent per month per tier.
- Incident response: P0 (security or attestation correctness): 15 minute acknowledgment, 4 hour resolution. P1 (degraded service): 1 hour acknowledgment, 8 hour resolution. P2 (minor): 24 hour acknowledgment, 5 day resolution.
- Status updates: every incident posted to status.dudilig.com within 15 minutes of detection.
- Post-mortems: P0 and P1 incidents publish a public post-mortem on status.dudilig.com within 5 business days.

18.3 Unit economics: per-action AI inference cost

Target ratios at Sonnet 4.6 / Opus 4.6 pricing as of mid-2026. Targets are aggressive but achievable with prompt-caching, batched inference where applicable, and routing of simple parses to Haiku-class models.

Action	Credits	Effective revenue (mid-tier)	Target AI inference cost	Target margin
Wallet sanctions screen	0.1	\$0.05	\$0.001 (mostly deterministic API)	98%
Investor KYC verification	2	\$1.00	\$0.10 (KYB orchestration + Claude synthesis)	90%
Smart contract attestation	5	\$3.00	\$0.50 (planner + tool calls + executable self-check)	83%
LP DDQ question	20	\$10	\$1.20 (RAG + multi-step reasoning + draft)	88%
Form D draft	50	\$30	\$3.00 (long-context generation)	90%
Auditor evidence packet	100	\$60	\$8.00 (large doc assembly)	87%
Multi-jurisdiction analysis	200	\$120	\$15.00 (cross-jurisdiction reasoning)	88%
Issuance Agent setup	350	\$200	\$25.00 (long session, multi-step)	88%

Target blended margin: 80 percent or higher at scale. Achieved by mix of high-margin volumetric actions (sanctions screens dominate B2B2C credit consumption). Per-action margin is tracked weekly and published internally.

18.4 Activation and engagement metrics

- **Time-to-first-credit-consumed (TTFCC).** Target: under 15 minutes from signup to first credit consumed in sandbox or paid tier. Measured continuously.
- **Time-to-first-attestation-shipped.** Target: under 30 minutes from signup to first attestation generated against a real contract.
- **Quickstart completion rate.** Percent of new sandbox signups that complete the quickstart in 10 minutes. Target: 70 percent.

- **Free-to-paid conversion.** Target: 17 percent (matches HIP-3 PLG modeled rate, validated against analogous developer-tool funnels like Vercel and Resend).
- **Weekly active credit-consuming customers.** Tracked as the engagement north-star.
- **Attestations per active customer per month.** Tracked for product-fit signal.
- **Trust Report public-share rate.** Percent of attestations whose Trust Report URL is shared externally. Tracked as the inbound flywheel signal.

18.5 Engineering velocity metrics

- **Deploy frequency.** Target: multiple deploys per day post-CTO and 2-engineer hire.
- **MTTR.** P0: under 4 hours. P1: under 8 hours. P2: under 24 hours.
- **Eval CI pass rate.** 100 percent required to merge to main. Failed evals block deploys.
- **Test coverage.** 80 percent or higher on agent orchestration code; 95 percent or higher on guardrails code; 100 percent on cryptographic signing pipeline.
- **Lead time for changes.** Target: under 24 hours from PR open to production for non-eval-affecting changes.

18.6 AI-native tooling stack (named choices)

Layer	Choice	Rationale
LLM provider (primary)	Anthropic Claude (Sonnet 4.6 + Opus 4.6)	Established choice; Sonnet for high-volume parsing, Opus for complex legal reasoning.
Model gateway	Portkey	Provider redundancy, cost monitoring, rate limit handling, request caching.
Eval framework	Inspect AI (Anthropic-developed, open-source)	Best fit for agent eval, native multi-turn support, schema-validated outputs.
LLM observability	Langfuse (self-hosted on Railway)	Open-source, multi-agent trace support, prompt versioning.
Prompt registry	Git-tracked YAML in monorepo	Versioned with code, simpler than dedicated tool at v1, supports CI-gated rollouts.
Vector DB	pgvector on Supabase	Already in stack, no new infrastructure at v1, scales to 10M+ vectors.

Feature flags	GrowthBook (self-hosted)	Open-source, self-hostable, no per-MAU pricing, supports model-version rollouts.
Synthetic data generation	Claude-driven pipeline + manual curation	Generates golden set candidates, humans hand-label final set.
APM and tracing	OpenTelemetry + Grafana Cloud	Standard stack; OTel works for LLM traces too via semantic conventions.
Error tracking	Sentry	Standard.
Status page	Statuspage.io (CNAME to status.dudilig.com)	Industry-standard expectation for paid customers.

Why these specific tools. Each choice is justified by self-hostability where possible (no per-MAU vendor lock), CI-friendliness (every change goes through eval gate), and total cost under \$2K per month at v1 scale. The stack is replaceable; the discipline of having a stack is not.

18.7 AI failure mode catalog

Every failure mode below has a defined detection mechanism, customer-facing behavior, operator alert, and recovery procedure. The catalog is the runbook.

Failure mode	Detection	Customer-facing behavior	Operator alert	Recovery
Hallucination in attestation	Adversarial probe set + customer flag	Verdict refused with explanation; no attestation issued	P0 page + incident	Route to human review queue; rollback model if pattern detected
Refusal when answer was warranted	Customer-reported + post-eval trend	Refusal message with "request human review" CTA	P2 ticket	Prompt tuning + eval addition for next CI cycle
Prompt injection in Solidity comment	Pre-ingestion sanitization + post-eval	Sanitized + flagged in attestation evidence	P3 log	Pattern added to sanitization rules; eval probe added
Schema violation on agent output	Structured output validator (Zod or Pydantic)	Retry up to 3 times with schema guidance	P2 ticket if persistent	Prompt revision + schema simplification

Cross-tenant prompt leak	Tenant-id assertion at retrieval boundary; embedding namespace check	Hard rejection of cross-tenant request	P0 page + incident	Incident report; tenant isolation audit
Context window overflow	Token counter pre-call	Chunking strategy invoked; explicit "too long, splitting analysis" message	P3 log	Rule pack chunking review
Model unavailable / API error	Provider response + Portkey health check	Retry with exponential backoff, then fallback model via Portkey	P1 page if sustained	Gateway routes to alternate; status page updated
Confidence below threshold	Model confidence score	Routed to human review queue with full context	P2 ticket	Compliance officer reviews and signs
Rule pack version mismatch	Pinning check at attestation time	Customer prompted to pin or upgrade	P3 log	UX nudge to pin to latest
RPC node unreachable	Alchemy/Quick Node health check	Retry then surface "blockchain data unavailable"	P1 page if sustained	Gateway falls back to alternate provider
Eval regression in CI	Inspect AI baseline diff	Deploy blocked	P1 ticket immediately	Investigation before any merge to main

18.8 Model regression gates and rollback policy

- **Major regression** (over 5 percent precision drop on any rule, or any hallucination found in production): immediate rollback to previous model version + incident report + customer notification within 24 hours if affected attestations.
- **Minor regression** (1 to 5 percent precision drop): rollback + investigation + fix within 7 days.
- **Eval CI gate.** Every prompt change, model version bump, or rule pack edit must pass the full Inspect AI eval suite before merge. No exceptions.

- **Customer-pinned attestations.** Existing customer attestations are pinned to (model_version, rulepack_version) and are not affected by upgrades (PRD section 10.2 reproducibility pinning). Customers explicitly opt-in to upgrades.
- **Model deprecation cadence.** When Anthropic or another provider deprecates a model version, Dudilig provides 90 days of notice to customers with pinned attestations on that version, plus a free re-attestation under the new pinned version.

19. Developer experience and customer-facing AI transparency (NEW in v1.2.3)

The product must feel like Stripe or Plaid or Resend to developers. That is not a marketing claim; it is a measurable set of surfaces. This section specifies them.

19.1 Public API surface

- **Protocol.** REST over HTTPS, JSON request and response bodies. No XML, no SOAP, no GraphQL at v1.
- **Authentication.** Two modes: API keys (Stripe pattern, sk_test_ for sandbox and sk_live_ for production), and OAuth2 (for B2B2C platform integrations that act on behalf of their end-users).
- **Specification.** OpenAPI 3.1 spec published at docs.dudilig.com/openapi.json. SDKs auto-generated from the spec.
- **Versioning.** URL-versioned (/v1/, /v2/). 12-month deprecation policy with 90-day customer notice.
- **Error format.** RFC 7807 Problem Details (type, title, status, detail, instance). Every error response includes a request ID for support.
- **Idempotency.** All POST endpoints accept an Idempotency-Key header (Stripe pattern). Replies return the original response within a 24-hour window.
- **Rate limits.** Sandbox: 100 req/min per key. Issuer tier: 1000 req/min. Boutique: 5000 req/min. Platform and above: custom. 429 Too Many Requests responses include Retry-After and X-RateLimit-Remaining headers.
- **Webhooks.** HMAC-SHA256 signed (Stripe pattern). Retry with exponential backoff over 24 hours (10 attempts: 1m, 5m, 15m, 30m, 1h, 2h, 4h, 8h, 12h, 24h). Webhook event catalog published at docs.dudilig.com/webhooks. Customers verify signatures with helper functions in every SDK.

19.2 SDKs

- **TypeScript SDK** (first, ships v1, days 31 to 60). Published to npm as @dudilig/sdk. Full v1 API coverage. Webhook signature verification helpers. Strongly typed (generated from OpenAPI). React hooks bundle for dashboard integration.
- **Python SDK** (second, days 91 to 180). Published to PyPI as dudilig. Full v1 API coverage. Async-first. Type-hinted (Pydantic models).

- **Go SDK** (deferred to v2).
- **Solidity helper library** (v2). Helper contract for embedding Trust Report verification in customer smart contracts.

Every SDK ships with: README quickstart, generated API reference, webhook verification helper, attestation helper, error types, retry helper, idempotency-key helper.

19.3 dudilig CLI

Single-binary CLI installable via `npm install -g @dudilig/cli` or `curl -fsSL install.dudilig.com | sh`. Commands:

`dudilig auth login` # interactive auth (browser flow)

`dudilig auth whoami` # show current key + tier

`dudilig attest <contract-address>` # run attestation

`dudilig attest --rulepack reg-d-506c <contract>`

`dudilig trust-report <issuer> <token>` # generate Trust Report

`dudilig trust-report --pretty <id>` # human-readable view of an existing report

`dudilig rulepack list` # list available rule packs

`dudilig rulepack show <id>` # inspect a rule pack

`dudilig rulepack diff <v1> <v2>` # compare versions

`dudilig audit-log replay <attestation-id>`

`dudilig sandbox start` # spin up local sandbox

`dudilig sandbox seed` # load demo contracts

`dudilig --version`

`dudilig --help`

CLI talks to the same REST API as the SDKs. Designed for power users, CI pipelines (compliance gates in deployment workflows), and the YC / Speedrun demo path ("install, attest a contract, get a Trust Report URL in under 60 seconds at the terminal").

19.4 Documentation site (docs.dudilig.com)

Structure modeled on stripe.com/docs and plaid.com/docs:

- **Quickstart.** Sub-10-minute path from signup to first attestation. Single page. Code samples in TypeScript, Python, curl, and CLI.

- **API reference.** Auto-generated from the OpenAPI spec. Every endpoint has request/response examples in 4 languages.
- **SDK guides.** Per language: install, authenticate, run attestation, verify webhook, handle errors.
- **Concept guides.** Rule packs, attestations, Trust Reports, agents, the determinism boundary, reproducibility pinning, the four pillars (linked to PRD section 4).
- **Webhook events catalog.** Every event type, payload schema, when it fires.
- **Error code reference.** Every error code, what it means, how to handle it.
- **Compliance reference.** Per-jurisdiction rule pack documentation. Reg D 506(c), Reg S, MiCA, ERC-1400, ERC-3643. Each rule pack has its own page with the rules it enforces, citations to regulatory text, and example contracts.
- **Recipes.** "Attest a Reg D 506(c) Polygon deployment in 5 minutes." "Generate an LP-shareable Trust Report URL." "Verify a webhook signature in TypeScript." "Run an ERC-3643 multi-contract attestation." "Embed a Trust Report badge on your issuer site."
- **Changelog.** Versioned, dated, every API change announced with deprecation timelines.
- **Status page link** to status.dudilig.com.

Hosted on Vercel. Source in a public GitHub repo (Markdown). External contributors can submit doc PRs.

19.5 Sandbox environment

- **No credit card required** to access. Resend-pattern signup: email + password, instant API key, immediate sandbox access.
- **100 free credits per month** in sandbox.
- **Pre-loaded demo contracts.** T-REX reference deployment on Polygon Mumbai, ERC-1400 reference on Sepolia, mock Reg D 506(c) Polygon deployment with intentional MAX_INVESTORS mismatch (the canonical demo contract).
- **Sandbox endpoints.** [sandbox.dudilig.com/v1/...](https://sandbox.dudilig.com/v1/) separate from production [api.dudilig.com/v1/...](https://api.dudilig.com/v1/) Sandbox responses include X-Dudilig-Mode: test header.
- **API key prefixes.** `sk_test_` and `sk_live_` (Stripe pattern). Visual differentiation in the dashboard. Test-mode visibility in every UI surface (banner, watermark on Trust Reports).
- **Sandbox-to-production migration.** One-click: same workspace, same rule packs, switch the key prefix. No re-onboarding.

19.6 Status page (status.dudilig.com)

- **Real-time uptime per service.** API, Attestation Engine, Investor Monitor, Filings Agent, Chat Agent, Trust Report, Webhooks, Sandbox.
- **Model availability.** Which Claude model versions are in production right now. Pinned versions per service.
- **Current rule pack versions** per jurisdiction.
- **P95 latency trend.** 24 hour, 7 day, 30 day rolling.
- **Eval pass rate trend.** Last 30 days.
- **Incident history** with public post-mortems for P0 and P1 incidents.
- **Subscription:** customers can subscribe to incident notifications by email, Slack, or webhook.

Hosted on Statuspage.io with CNAME to status.dudilig.com.

19.7 Customer-facing reasoning transparency

Every attestation page includes:

- **Step-by-step agent reasoning trace.** Planner decomposition, specialist tool calls, executable self-check, synthesis. Each step visible with input, tool used, output.
- **Tool calls** with inputs and outputs. Customer sees that the OFAC check called `chainalysis.screen(0xabc...)` and got `{match: false}`, that the AST parser called `parseSolidity(source)` and returned the AST, that the rule pack invocation called `applyRulePack('reg-d-506c-v3.2', ast)` and returned the verdict array.
- **Per-rule confidence scores** (0 to 100). Shown as badges next to each Pass/Fail/Warning.
- **Citations** to specific Solidity lines or rule pack sections for every verdict.
- **Model card per attestation.** "This attestation was generated by claude-sonnet-4-6 against rulepack-reg-d-506c-v3.2 at block 19,847,221 on May 1, 2026. Reproducibility pin: attestation-id-0xabc123."
- **Replay button** (Auditor mode). Re-runs the attestation with the pinned model and rule pack and produces a byte-identical result.

Eval transparency dashboard at trust.dudilig.com/evals (publicly accessible):

- Per-rule precision and recall, last 30 days, all customers (anonymized in aggregate).
- Hallucination rate trend.
- Multi-turn ASR trend.

- Eval suite version and last update date.

This is the single highest-credibility move available to an AI-native compliance product. Vanta does not show its work; Dudilig does. Showing the work is the wedge.

19.8 Open-source strategy

Open-source under Apache 2.0 at github.com/dudilig:

- Rule packs (Reg D 506(c) first, then Reg S, MiCA, ERC-3643, ERC-1400). Community contributions accepted with CLA.
- Evaluation harness: the Inspect AI eval scaffolding plus a curated public golden contract set (subset of the full set; full set stays internal).
- TypeScript SDK source.
- dudilig CLI source.
- OpenAPI specification.
- Webhook verification helpers.

Closed source (the moat):

- The agentic orchestration layer (planner, specialist coordination, executable self-check implementation).
- The mismatch corpus (the gold dataset, PRD section 4.2.2).
- KYC decision trails (PRD section 4.2.4).
- Customer-specific golden sets (pinned to the customer).
- The model fine-tuning pipeline.
- The Trust Report cryptographic signing pipeline.
- The closed-source bytecode analysis path (productized from IC3 Decompiling-with-LLM paper).

Why this split. Open-sourcing the rule packs creates community-driven rule pack improvement, builds developer mindshare, and lets customers verify what Dudilig is actually checking. Open-sourcing the SDK and CLI lowers integration friction. Closed-sourcing the orchestration and the data corpora preserves the data moat (PRD Pillar 2). This is the same split Vanta uses for its security questionnaire framework versus its attestation engine.

20. Citations and sources

- RWA.xyz public dashboard (rwa.xyz). On-chain RWA total value tracking, accessed Q1 2026.

- McKinsey and Company, "From ripples to waves: The transformational power of tokenizing assets." June 2024.
- BCG and ADDX, "Relevance of On-Chain Asset Tokenization in Times of Crypto Winter." September 2022.
- Standard Chartered, "Real-World Asset Tokenization: From Ripples to Waves." 2024.
- US SEC, Regulation D Rule 506(c). Securities Act of 1933, as amended.
- Tokeny, ERC-3643 standard documentation (erc3643.org and tokeny.com).
- Initiative for Cryptocurrencies and Contracts (ic3research.org). Co-headquartered at Cornell Tech. Ari Juels, co-director.
- IC3 papers cited in section 11. Arxiv preprint IDs as held in the Dudilig PRD Thread Export. To be re-verified against live arxiv listings before any external circulation.
- Vanta public pricing and Trust Center reference (vanta.com).
- Federal Reserve / OCC SR 11-7, "Guidance on Model Risk Management."
- NYDFS 23 NYCRR Part 504, "Banking Division Transaction Monitoring and Filtering Program Requirements."
- FFIEC IT Examination Handbook, Vendor Management section.
- European Union Artificial Intelligence Act (Regulation EU 2024/1689). EU AI Act high-risk system classification under Annex III. Conformity assessment requirements under Articles 16 to 29.
- Colorado AI Act (Senate Bill 24-205), effective February 1, 2026.
- General Data Protection Regulation (Regulation EU 2016/679), Article 22 on automated individual decision-making.
- Anthropic Claude API documentation, including Sonnet 4.6 and Opus 4.6 model cards.
- Inspect AI evaluation framework (open-source, github.com/UKGovernmentBEIS/inspect_ai), used for the eval suite scaffolding.
- Stripe API documentation as a developer-experience benchmark for `sk_test_`/`sk_live_`, idempotency keys, RFC 7807 error format, and HMAC-signed webhooks.
- Plaid developer documentation as a benchmark for financial-data API surface design.
- Resend developer onboarding flow as a benchmark for sandbox-first, no-credit-card developer signup.

- Securitize Holdings, Inc. EDGAR filings (CIK 0002094496) and Securitize, Inc. EDGAR filings (CIK 0001762096), including S-4/A Amendment No. 1 (accession 0001213900-26-042685, filed April 13, 2026).
- Ethereum Attestation Service documentation (attest.org and github.com/ethereum-attestation-service). Schema registry and attestation contract architecture; EAS Scan public verification surface; EIP-712 signing; off-chain and Merkle-tree private data modes.
- Quicknode developer guide, "What Is Ethereum Attestation Service (EAS) & How to Use It," updated November 2025. Reference for schema creation, attestation issuance, and SDK integration patterns used in section 10.5.
- Chainlink, "Compliance Attestation: A Comprehensive Guide," updated February 2026. Reference for the Automated Compliance Engine (ACE), Proof of Reserve, Chainlink Runtime Environment (CRE), and Confidential Compute used in section 12.
- Tokeny and Apex Group, "T-REX Ledger" launch announcement, March 2026. Reference for the cross-chain compliance state layer positioning used in section 12.1.
- Dudilig Bundled-Platforms Competitive Brief (May 2026, internal). Tier 1 mapping reference.
- Dudilig Infrastructure-Layer Competitive Brief (May 2026, internal). Tier 2 mapping and EAS substrate posture reference.

End of Dudilig PRD v1.2.4.